Abstract

This thesis examines whether Bitcoin prediction market prices are consistent with a probability benchmark constructed from information available in cryptocurrency derivatives markets. The analysis combines historical observations from Polymarket and Kalshi with Bitcoin spot prices from Binance and the Deribit DVOL index. Benchmark probabilities are obtained using a lognormal terminal-price model, with spot and volatility inputs matched according to their availability at each observation timestamp to avoid look-ahead bias.

The principal analysis focuses on 19,649 observations from 2,847 Polymarket terminal markets. On average across markets, Polymarket probabilities exceed the DVOL-based benchmark by approximately 0.75 percentage points. The market-level mean signed mispricing is 0.007496, with a 95% bootstrap confidence interval of [0.006709, 0.008285]. The estimate remains positive under monthly aggregation and is almost unchanged under a seven-day maturity restriction, although the latter removes only 33 observations. Larger average differences are observed for Kalshi and for path-dependent Polymarket contracts. These estimates are interpreted separately because of differences in contract design, trading environment, and payoff comparability with a terminal benchmark.

The findings document systematic cross-market pricing differences but do not establish forecasting errors, causal relationships, or executable arbitrage opportunities. The thesis contributes a transparent framework for comparing prediction market valuations with a derivatives-based reference while accounting for information availability, payoff structure, and dependence among observations.

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1 Introduction

Prediction markets allow participants to trade contracts whose payoffs depend on the occurrence of future events. Their prices aggregate information, expectations, and trading preferences into a single market valuation. For binary contracts, prices are often interpreted as probabilities, although this interpretation depends on assumptions about risk preferences, market frictions, and the relationship between prices and beliefs (Manski 2006; Wolfers and Zitzewitz 2006). A prediction market price is therefore best understood as a probability-like valuation generated within a particular trading environment, rather than as a direct observation of an objective probability.

When the event concerns the future price of a financial asset, prediction market valuations can be compared with information from derivatives markets. Option prices reflect market assessments of the distribution of future asset prices and can be used to construct risk-neutral or risk-neutral-style probabilities for specified terminal events. Such comparisons offer a way to examine whether economically related claims receive similar valuations across markets. They do not, however, provide a mechanical test of market efficiency. Prediction markets and derivatives markets differ in participation, liquidity, contract design, risk exposures, and trading costs, while option-implied probabilities need not coincide with physical probabilities.

Bitcoin provides a useful setting for studying this question. It trades continuously in spot markets, supports an active derivatives market, and is the underlying asset for prediction contracts written on future price events. A contract paying one dollar if Bitcoin closes above a threshold can therefore be compared with a probability proxy constructed from contemporaneously available spot and implied volatility information. The comparison is less direct for contracts whose payoff depends on whether a threshold is reached before expiration, since these contracts depend on the price path rather than only on the terminal value.

The research question is therefore: to what extent are Bitcoin prediction market prices consistent with a derivatives-based probability benchmark, and do the observed differences vary across payoff structures and market environments? The analysis focuses primarily on terminal Polymarket contracts. Path-dependent Polymarket contracts are retained as a separate sample, while Kalshi provides a supplementary comparison with a regulated prediction market operating under a different institutional structure.

The analysis is related to recent work comparing cryptocurrency prediction markets with derivatives-based valuations (Valery 2025; Fabi et al. 2026). It adopts a more parsimonious benchmark than a complete reconstruction of the option-implied terminal distribution. Bitcoin spot prices and the Deribit DVOL index are combined in a lognor-

mal terminal-price model, producing a transparent probability proxy for each contract threshold and maturity. The purpose is not to replicate the full information contained in the option surface, but to construct a consistent market-based reference that can be applied across a large collection of contracts.

The dataset combines historical observations from Polymarket and Kalshi with Bitcoin spot data from Binance and DVOL observations from Deribit. The principal sample contains 19,649 observations from 2,847 Polymarket terminal markets. A further 10,323 observations from 584 path-dependent Polymarket markets are analysed separately. The Kalshi baseline sample contains 3,135 contracts associated with 485 daily events. Benchmark inputs are matched according to the information available at each observation timestamp, avoiding the use of future spot or volatility observations.

Signed mispricing is defined as the prediction market probability minus the corresponding DVOL-based benchmark probability. Positive values mean that the prediction market assigns a higher valuation to the event than the benchmark; negative values indicate the opposite. Because the term is used as an empirical measure of a pricing difference, non-zero mispricing does not by itself establish an executable arbitrage opportunity, a forecasting error, or a failure of market efficiency. The analysis reports both signed and absolute differences and uses inference procedures that account for dependence among observations from the same market or event.

The main result is a small but precisely estimated positive difference for Polymarket terminal contracts. The market-level mean signed mispricing is 0.007496, with a 95% bootstrap confidence interval of [0.006709, 0.008285]. On average across markets, Polymarket terminal probabilities therefore exceed the benchmark by approximately 0.75 percentage points. The mean absolute difference is larger, at approximately 2.21 percentage points, because positive and negative deviations partly offset each other in the signed mean. The result remains positive when observations are aggregated by month and when the sample is restricted to contracts with no more than seven days to maturity, although the latter restriction removes only a small number of observations.

The supplementary samples produce larger average differences. Kalshi has an event-level mean signed mispricing of 0.070670 under the 30-minute baseline selection rule. Path-dependent Polymarket contracts have a market-level mean of 0.076566. Neither estimate should be read as a direct ranking of market efficiency. Kalshi contracts differ from the main Polymarket sample in horizon, contract organisation, and trading environment. For path-dependent contracts, the use of a terminal benchmark creates an additional mismatch between the payoff being valued and the probability model.

Regression estimates describe how signed mispricing varies with the benchmark probability, time to maturity, DVOL, contract type, and market environment. They are treated

as conditional associations rather than causal effects. The unified specification associates Polymarket terminal observations with lower signed mispricing than Kalshi observations, while the path-dependent indicator captures an additional difference relative to Polymarket terminal contracts. The benchmark probability coefficient is interpreted cautiously because the same probability also enters the construction of the dependent variable.

The contribution of the thesis is therefore empirical and methodological. It provides a transparent cross-market comparison, keeps terminal and path-dependent claims separate, aligns benchmark inputs with their actual availability, and conducts inference at the level where observations may be dependent. The findings document systematic differences between prediction market prices and a DVOL-based reference, but their economic interpretation remains conditional on the benchmark model and on the institutional differences between the markets.

Chapter 2 develops the theoretical connection between prediction market contracts, implied probabilities, and event-contingent claims. Chapter 3 reviews the literature on prediction market efficiency, market frictions, modern prediction platforms, and derivatives-implied information. Chapters 4 and 5 describe the data, sample construction, benchmark model, and empirical methods. Chapter 6 presents the results, and Chapter 7 discusses their economic interpretation, limitations, and implications.

2 Prediction Markets and Event-Contingent Claims

This chapter develops the theoretical and institutional background required to interpret prediction market prices as event-contingent claims and compare them with probability benchmarks derived from financial derivatives. The aim is not to provide a comprehensive history of prediction markets. The discussion instead focuses on the concepts needed for the empirical analysis conducted in this thesis.

The chapter first considers prediction markets as mechanisms for aggregating dispersed information and then examines the conditions under which a binary contract price can be interpreted as a market-implied probability. The connection between these contracts and financial derivatives is developed through event-contingent claims, digital options, and cash-or-nothing payoffs. The remaining subsections explain why Bitcoin is well suited to a cross-market comparison and introduce Polymarket and Kalshi, the two prediction market environments examined in the empirical analysis.

2.1 Prediction Markets as Information Aggregation Mechanisms

Prediction markets are trading environments in which participants exchange contracts whose payoff depends on a future event. These events may concern elections, macroeconomic announcements, corporate decisions, sporting results, technological developments, geopolitical events, or financial variables. The present analysis focuses on contracts linked to the future price of Bitcoin, such as whether it will finish above a specified threshold or within a given range on a future date.

The main economic rationale for prediction markets is that trading can bring together information held by different individuals. Participants may observe different private signals, interpret public information differently, or assign different probabilities to the same outcome. By buying and selling event-contingent claims, they commit capital to these assessments and allow them to influence a common market price.

This mechanism is related to the broader view of prices as tools for aggregating dispersed information (Hayek 1945). A prediction market price is not simply the average belief of everyone participating in the market. It is determined at the margin by traders who are willing to transact at prevailing prices. The information it contains may therefore differ from the information collected through a survey or opinion poll.

A survey records the stated beliefs of respondents, whereas a market price reflects beliefs that participants are prepared to support through actual trading decisions. Participation in a prediction market is voluntary, traders need not represent the wider population, and their decisions have financial consequences.

These incentives are part of the aggregation process. Traders with better information,

more accurate models, or a stronger interpretation of public signals may trade when they believe that the current price is incorrect. Their orders can move the price towards a level that incorporates their information. Participants whose assessments are persistently inaccurate may incur losses and gradually exercise less influence on price formation.

None of this guarantees that prediction markets produce efficient or unbiased forecasts. Their performance depends on the conditions under which trading takes place. Limited liquidity, participation constraints, heterogeneous risk preferences, transaction costs, behavioural demand, and market segmentation may prevent prices from reflecting all available information. Prediction market prices are therefore better viewed as market-based assessments of future events than as mechanical estimates of objective probabilities.

This qualification is relevant to the empirical analysis. A difference between a prediction market price and a benchmark constructed from another market does not by itself establish irrationality or identify an arbitrage opportunity. It shows that two trading environments assign different probability-like valuations to economically related events defined on the same underlying asset. Interpreting that difference requires attention to the institutional and financial conditions under which both prices are formed.

The payoff structure of a binary contract provides a more precise basis for its probability interpretation. It also explains why prediction market prices can be compared with the derivatives-based benchmark used later in the thesis.

2.2 Prices as Implied Probabilities

The probability interpretation of a prediction market price starts from the payoff of a binary event contract. Consider a contract linked to an event A. It pays one monetary unit if the event occurs and zero otherwise:

$$X_A = \begin{cases} 1, & \text{if event } A \text{ occurs,} \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

Suppose that the contract trades at a price p, where

$$0 \leq p \leq 1. \tag{2}$$

Under suitable assumptions, p can be interpreted as the probability that market participants assign to event A. Consider, for example, a contract that pays one dollar if Bitcoin finishes above a specified threshold at expiration. A price of 0.60 corresponds approximately to a market-implied probability of 60 percent, while the complementary outcome

has a probability-like value close to 40 percent. This simple interpretation abstracts from discounting, transaction fees, bid–ask spreads, and other trading frictions.

Financial derivatives give rise to a related, but conceptually different, probability measure. For a replicable binary claim traded in an arbitrage-free market, risk-neutral valuation implies (Hull 2022)

$$V_t(A) = e^{-r\tau}\mathbb{E}_t^{\mathbb{Q}}\left[\mathbf{1}_A\right] = e^{-r\tau}\mathbb{Q}_t(A), \tag{3}$$

where $V_t(A)$ is the value of the claim at time t, r is the risk-free rate, and τ is the remaining time to settlement. The indicator $\mathbf{1}_A$ equals one when event A occurs, while $\mathbb{Q}_t(A)$ denotes its probability under the risk-neutral pricing measure. When discounting is negligible,

$$V_t(A) \approx \mathbb{Q}_t(A). \tag{4}$$

This relationship provides a financial benchmark for a binary payoff. It does not imply that an observed prediction market price necessarily represents a unique risk-neutral probability. Prediction market prices emerge from trading among participants who may differ in beliefs, preferences, constraints, and information sets. The conditions under which these prices approximate aggregate beliefs are therefore different from the no-arbitrage assumptions used in derivatives pricing (Manski 2006; Wolfers and Zitzewitz 2006).

Interpreting a prediction market price as a probability also requires a clearly specified payoff and economically meaningful trading. Limited liquidity, bid–ask spreads, capital constraints, risk preferences, behavioural demand, and other market frictions can create a gap between the observed contract price and the physical probability of the event. The market price should not automatically be treated as an unbiased estimate of the corresponding real-world probability.

The distinction becomes particularly relevant in a cross-market comparison. A probability derived from financial derivatives is generally associated with a risk-neutral pricing measure. A prediction market price may reflect subjective beliefs together with risk preferences and market-specific frictions. A difference between the two can therefore have several economic explanations and cannot be classified directly as a forecasting error.

The empirical analysis treats prediction market prices as probability-like market valuations. It does not attempt to determine which market reveals a uniquely correct probability. The narrower question is whether comparable Bitcoin events receive systematically different valuations across market environments.

The next subsection places binary prediction market contracts within the broader framework of event-contingent claims and financial derivatives.

2.3 Event-Contingent Claims and Financial Derivatives

Prediction market contracts belong to the broader class of event-contingent claims, whose payoff depends on the realisation of a specified future state or set of states. In the simplest binary contract, the holder receives one unit if the relevant event occurs and nothing otherwise.

This payoff is related to an Arrow–Debreu security (Arrow 1964). A pure Arrow–Debreu security pays one unit in a particular state of the world and zero in every other state. A binary event contract is generally defined over a broader event rather than a single elementary state. It can be understood as an aggregation of the state-contingent securities associated with all states in which the event occurs.

Prediction market contracts can therefore be placed within the framework of contingent-claim pricing. Their value depends on possible future outcomes and on the payment attached to those outcomes. Their economic structure resembles that of conventional derivatives, even though the two markets may differ in participation, trading arrangements, collateral, and settlement procedures.

The closest financial counterpart is a binary, or digital, option (Hull 2022). A cash-or-nothing digital call pays a fixed amount when the underlying asset finishes above a strike price K at expiration and zero otherwise. For a unit payoff,

$$V_T^{\text{call}} = \mathbf{1}_{\{S_T > K\}} = \begin{cases} 1, & \text{if } S_T > K, \\ 0, & \text{otherwise,} \end{cases} \tag{5}$$

where S_T is the value of the underlying asset at expiration.

A prediction market contract asking whether Bitcoin will finish above K has the same basic terminal payoff. A corresponding lower-tail contract resembles a cash-or-nothing digital put:

$$V_T^{\text{put}} = \mathbf{1}_{\{S_T < K\}}. \tag{6}$$

The same reasoning can be applied to range contracts. Consider a claim that pays one unit when the terminal Bitcoin price lies between a lower boundary K_L and an upper boundary K_U:

$$K_L < S_T < K_U. \tag{7}$$

Its payoff is

$$V_T^{\text{range}} = \mathbf{1}_{\{K_L < S_T < K_U\}}. \tag{8}$$

If the terminal price follows a continuous distribution, the probability assigned to the interval can be written as

$$\Pr(K_L < S_T < K_U) = \Pr(S_T < K_U) - \Pr(S_T \leq K_L). \tag{9}$$

A range contract can thus be represented through the difference between two digital claims, allowing the valuation to be associated with the probability mass between the two boundaries.

This digital-option analogy provides the financial basis for the benchmark used in the thesis. When a prediction market contract and the derivatives-market information refer to the same underlying asset, comparable thresholds, and compatible horizons, the derivatives market can provide an external valuation of the relevant event.

The comparison is necessarily imperfect. Conventional derivatives exchanges often include professional market makers, standardised margin and collateral arrangements, broader hedging opportunities, and substantial institutional participation. Prediction markets may have a different investor base, lower liquidity, other trading constraints, and distinct settlement rules. Prices can differ even when the underlying economic event appears similar.

The timing of the payoff introduces another distinction. Some prediction market contracts depend only on the Bitcoin price at expiration. These terminal claims are directly comparable, in payoff terms, with European-style digital options and with probabilities calculated from a terminal price distribution.

Other contracts pay when Bitcoin reaches a specified threshold at any point before expiration. Their payoff depends on the path followed by the underlying, not only on S_T. These claims are closer to barrier or touch options and require a different theoretical treatment (Hull 2022).

Whether a contract is terminal or path-dependent determines how meaningfully the benchmark can be applied. The distinction is therefore part of the empirical design, rather than a purely descriptive classification. Terminal contracts form the main com-

parison sample, while path-dependent contracts are analysed separately.

Viewing prediction market contracts as contingent claims explains why derivatives-pricing methods can be used to study their prices, while also defining the limits of that comparison. An empirical application requires an asset for which prediction contracts, observable spot prices, and a developed derivatives market coexist. Bitcoin meets these requirements.

2.4 Bitcoin as a Natural Empirical Setting

Bitcoin offers a useful setting for comparing prediction market probabilities with valuations derived from financial derivatives. The same underlying asset is observed in several distinct trading environments: it trades continuously in spot markets, supports an active derivatives market, and is the subject of numerous prediction market contracts.

This combination is less common in other prediction market applications. Elections, geopolitical developments, and technological outcomes may attract considerable trading activity, but they generally lack a parallel derivatives market from which an independent market-based probability benchmark can be constructed. Bitcoin price events, by contrast, can be defined as contingent claims on an observable financial asset.

A prediction market may, for example, offer a contract on whether Bitcoin will finish above a given threshold on a future date. At the same time, Bitcoin option prices contain information about the distribution that derivatives-market participants assign to the future value of the asset. The coexistence of these markets makes a cross-market comparison of probability-like valuations possible.

Recent research has begun to study this relationship by comparing cryptocurrency prediction market contracts with valuations obtained from options and other derivatives (Valery 2025; Fabi et al. 2026). This literature connects the pricing of binary event contracts with information contained in cryptocurrency derivatives. Its findings and its relationship to the present thesis are considered in the Literature Review.

Deribit operates an active Bitcoin options market and publishes the Bitcoin DVOL index. DVOL summarises information from Bitcoin option prices as a rolling 30-day annualised measure of implied volatility (Fernando 2021). In this thesis, it is used as an aggregate volatility input in the construction of the derivatives-based probability benchmark. It does not replace a complete volatility surface, but it offers a transparent and consistently available proxy for the volatility implied by the options market.

Bitcoin's continuous trading is also useful for the empirical design. Spot and derivatives markets operate twenty-four hours a day, seven days a week, so the underlying price remains observable throughout the life of a prediction market contract. This differs from assets traded only during conventional exchange hours and reduces ambiguity about the

value of the underlying at a given observation time.

Continuous trading does not mean that all venues incorporate information at the same speed. Prediction markets and derivatives markets can differ in liquidity, investor composition, trading activity, and market microstructure. Participants in both markets nevertheless respond to developments in the same underlying asset, making the resulting valuations economically comparable even when their institutional settings differ.

Bitcoin's high volatility also makes threshold and range contracts economically relevant. Prediction market participants can trade views about specific future price events, while the options market reflects volatility embedded in derivatives prices. The two markets therefore encode information about future Bitcoin prices in different forms.

The purpose of the analysis is not to forecast Bitcoin returns or identify the most accurate forecasting model. It asks a narrower question: do distinct market environments assign consistent probability-like valuations to comparable future Bitcoin price events?

The setting also allows the role of market segmentation to be considered. Prediction market participants and cryptocurrency derivatives traders need not belong to the same investor population. They may differ in sophistication, capital, information, trading objectives, and hedging opportunities. Prices for related Bitcoin events can consequently be formed under substantially different institutional conditions.

For the empirical analysis, Bitcoin offers a continuously observable underlying asset, actively traded event-contingent contracts, and a developed derivatives market from which an external benchmark can be constructed. The following subsection introduces Polymarket and Kalshi and discusses the institutional differences most relevant to interpreting their prices.

2.5 Polymarket and Kalshi

The empirical analysis uses data from Polymarket and Kalshi. Both platforms offer contracts linked to future Bitcoin prices, but their institutional designs differ considerably. Including them allows the analysis to examine probability-like valuations formed in two distinct trading environments.

Institutional design can influence price formation through participation, liquidity, order execution, collateral arrangements, settlement rules, and regulation. These differences must therefore be considered when interpreting the observed probabilities on each platform.

The Polymarket data used in this thesis come from its hybrid-decentralised central limit order book. Orders are matched off-chain, while settlement takes place on-chain through

smart contracts operating on Polygon (Polymarket 2026). Prices are determined by orders submitted by market participants rather than by a bookmaker or centralised forecaster.

Polymarket offers Bitcoin-related contracts with identifiable price thresholds, expiration conditions, and historical price observations. Some depend on the Bitcoin price at expiration, while others resolve when a threshold is reached during a specified period. As discussed above, this difference in payoff structure determines how the contracts enter the empirical analysis.

Polymarket provides the main prediction market sample. Its Bitcoin contracts vary in thresholds, horizons, and payoff structures, and its terminal contracts can be compared directly with probabilities calculated from the derivatives-based terminal benchmark.

Kalshi represents a different institutional setting. KalshiEX operates as a Designated Contract Market under the oversight of the U.S. Commodity Futures Trading Commission (Commodity Futures Trading Commission 2020). Its event contracts are traded within a regulated framework that differs from Polymarket's hybrid blockchain-based architecture.

Kalshi is used as a supplementary comparison. Its inclusion makes it possible to examine Bitcoin-related probability valuations formed under another regulatory structure, contract design, and trading environment. Evidence from both platforms gives a broader view of cross-market differences, although it does not create a perfectly matched comparison.

The two datasets are not mechanically equivalent. Their Bitcoin contracts differ in effective horizon, payoff specification, frequency, and trading conditions, and a one-to-one match across platforms is generally unavailable. These differences limit direct comparisons of their average deviations from the benchmark.

Polymarket terminal contracts consequently form the principal empirical sample, while Kalshi is analysed as a supplementary regulated-market comparison. This treatment preserves the institutional dimension of the analysis without assuming that the platforms offer identical financial claims.

Market-implied probabilities are formed within particular institutional arrangements. Differences in participation, liquidity, collateral requirements, trading technology, settlement, and regulation may all influence the valuations observed in each market.

These considerations define the framework used in the empirical analysis. Prediction market contracts can be viewed as event-contingent claims, and their prices can be treated as probability-like valuations under appropriate assumptions. Bitcoin supplies a common underlying asset across prediction, spot, and derivatives markets. Polymarket and Kalshi then provide two distinct settings in which probabilities for Bitcoin-related events emerge through trading.

The next chapter reviews the literature on prediction market efficiency, the interpretation of binary contract prices, behavioural and market-friction explanations for pricing differences, cryptocurrency prediction markets, and the use of derivatives-implied information as a benchmark for event probabilities.

3 Literature Review

This chapter reviews the literature most closely related to the empirical analysis conducted in this thesis. Chapter 2 developed the theoretical interpretation of prediction market contracts and their connection with financial derivatives. The focus here is instead on the empirical evidence concerning prediction market efficiency, forecasting accuracy, pricing frictions, and cross-market differences.

Early studies examine whether prediction markets aggregate dispersed information and produce accurate forecasts, while a related strand documents behavioural biases, liquidity constraints, and other frictions that may affect observed prices. More recent work studies modern platforms such as Polymarket and Kalshi, with particular attention to calibration, trader heterogeneity, liquidity, and market microstructure. A separate literature uses option prices to recover information about the future distribution of financial assets. Bringing these strands together, recent studies have begun to compare cryptocurrency prediction market prices directly with valuations derived from derivatives markets.

These different contributions provide the background for the empirical question examined in this thesis. Evidence on prediction market efficiency motivates the use of contract prices as informative market valuations, while the literature on market frictions gives reasons why they may differ from an external benchmark. Research on option-implied information provides the financial basis for constructing such a benchmark. The final part of the chapter places the present analysis within this emerging cross-market literature and sets out the empirical expectations that guide the analysis.

3.1 Prediction Market Efficiency and Forecast Accuracy

A substantial part of the prediction market literature examines whether market prices provide useful forecasts of future events. The main empirical question is whether the information revealed through trading produces estimates that are at least as accurate as alternative forecasting methods, such as opinion polls, expert assessments, or statistical models. The evidence is generally favourable, although performance varies across markets, horizons, and contract designs.

Early studies focused mainly on political prediction markets. Forsythe et al. (1992) analyse the Iowa Political Stock Market during the 1988 U.S. presidential election. Despite the small and non-representative set of participants, market prices produced forecasts that compared favourably with contemporaneous opinion polls. The authors also show that average trader beliefs need not be unbiased for prices to remain informative. Their explanation rests on the role of marginal traders: average beliefs may exhibit substantial judgement bias, while prices are influenced more directly by participants trading at the

margin, who were found to be less affected by those biases.

Longer-run evidence from the Iowa Electronic Markets reaches a similar conclusion. Berg et al. (2008) compare prediction market prices with 964 election polls collected between 1988 and 2004. The market forecast was closer to the eventual outcome than the corresponding poll in approximately 74% of the comparisons. Its relative advantage was particularly clear at longer horizons, when polls contained less information about the final result.

Wolfers and Zitzewitz (2004) review evidence from political, economic, and corporate prediction markets. They conclude that these markets often aggregate dispersed information effectively and can outperform moderately sophisticated forecasting benchmarks. Prices also tend to react quickly when new public information becomes available. Their review does not imply that every prediction market is efficient, however. Forecast quality depends on participation, incentives, liquidity, contract design, and the information available to traders.

Snowberg et al. (2013) extend this discussion to economic forecasting. Their survey describes prediction markets as tools for collecting and combining beliefs about uncertain outcomes. Across several applications, market forecasts are often competitive with professional forecasts and surveys. The authors nevertheless emphasise that market design and institutional conditions affect the quality of the resulting predictions.

These findings require some conceptual care. Forecast accuracy, calibration, and informational efficiency are related but distinct properties. Forecast accuracy evaluates predictive performance against realised outcomes using a loss or scoring criterion appropriate to the type of forecast. Calibration considers whether events assigned a given probability occur with approximately that frequency across a sufficiently large set of contracts. Informational efficiency instead concerns whether market prices incorporate the information available to participants.

A market can therefore produce an accurate forecast without being fully informationally efficient, while an efficient price may still be followed by an outcome that was assigned a low ex ante probability. Neither forecast accuracy nor informational efficiency is equivalent to the absence of arbitrage, which imposes a separate set of conditions on prices, payoffs, and feasible trading strategies.

The literature consequently supports treating prediction market prices as informative market-based forecasts, but not as error-free estimates of physical probabilities. Evidence from election markets is particularly extensive, while its generalisation to financial and cryptocurrency events is less direct. Participant populations, trading technologies, liquidity conditions, contract design, and settlement procedures can differ substantially

across market environments.

The present thesis does not evaluate forecast accuracy by comparing prediction market prices with realised binary outcomes. Instead, it asks whether probability-like valuations for Bitcoin events are consistent with a benchmark constructed from derivatives-market information. The empirical question is therefore one of cross-market pricing consistency rather than a direct comparison of forecasting models.

These qualifications motivate the literature on behavioural biases, liquidity constraints, market frictions, and other limits to prediction market efficiency.

3.2 Behavioural Biases, Market Frictions, and Limits to Efficiency

The favourable evidence on forecast accuracy does not imply that prediction market prices are always unbiased or fully efficient. Prices emerge from the interaction of traders with different beliefs, preferences, budgets, and information. They are also shaped by the institutional conditions under which trading takes place. Systematic deviations can therefore coexist with prices that remain informative overall.

One qualification concerns the interpretation of a contract price as a probability. Manski (2006) shows that, with risk-neutral price-taking traders, heterogeneous beliefs, and limited trading budgets, the equilibrium price need not equal the average belief held by market participants. Instead, it corresponds to a particular quantile of the budget-weighted distribution of beliefs and only partially identifies their central tendency. The relation between price and aggregate beliefs therefore depends on the assumptions governing trading behaviour.

Wolfers and Zitzewitz (2006) reach a less restrictive conclusion under alternative assumptions. They derive sufficient conditions under which prediction market prices coincide with mean beliefs and show that, across a broader class of models, prices are often close to the average beliefs of traders even when exact equality does not hold. Risk aversion and the distribution of beliefs are central to this relationship. Read together, the two studies do not provide a simple yes-or-no answer to the probability interpretation of prediction market prices. They identify conditions under which that interpretation is more or less informative.

Empirical research also documents departures from perfect calibration. Page and Clemen (2013) find that prediction market prices become better calibrated as contract expiration approaches, while longer horizons display a more pronounced favourite-longshot pattern. Low-probability events tend to be overpriced and high-probability events underpriced, with the distortion increasing as the remaining time to expiration becomes longer.

The favourite-longshot pattern does not require a single behavioural explanation. Ottaviani and Sørensen (2010) show that its direction and magnitude can arise from the interaction between private information and noise in parimutuel prediction settings. Observable calibration biases should therefore not automatically be attributed to a particular form of irrationality or to identical preferences across traders.

Individual mistakes also need not translate directly into inefficient market prices. Oliven and Rietz (2004) examine trader behaviour in the Iowa Electronic Markets and find substantial differences between participants who set prices and those who mainly accept existing prices. Market-making traders were less prone to mistakes and appeared more rational than price-taking traders. Prices are determined by marginal trading decisions rather than by the average level of sophistication in the participant population, allowing informative market outcomes to coexist with systematic errors at the individual level.

Liquidity and trading frictions impose a separate constraint on information aggregation. In thin markets, informed participants may be unable or unwilling to trade enough to move prices towards their valuations. Wide bid–ask spreads can make transaction prices noisy measures of the current market assessment, while limited depth increases the price impact of individual orders. Transaction costs, funding constraints, position limits, and capital committed until settlement may further reduce the incentive to trade on relatively small pricing differences.

These frictions also limit the interpretation of apparent arbitrage opportunities. A difference between a prediction market price and an external benchmark does not imply the existence of a feasible risk-free strategy once trading costs, execution conditions, and differences in payoff or timing are taken into account.

Manipulation raises a related concern, particularly in markets with limited participation. Experimental evidence is more nuanced. Hanson et al. (2006) introduce traders with explicit incentives to distort prices and find that their attempts do not significantly reduce price accuracy. Participants without manipulation incentives respond by accepting trades at systematically different price levels, offsetting much of the attempted distortion. The result is specific to the experimental setting, however, and does not imply that prediction markets are generally immune to manipulation.

Market segmentation provides another reason why valuations may differ across venues. Prediction market participants and derivatives traders need not share the same information, objectives, risk-bearing capacity, or hedging opportunities. Economically related claims may consequently be priced under different information sets and institutional constraints. Cross-market disagreement alone is therefore insufficient to establish that one venue is inefficient.

The literature presents a qualified view of prediction market efficiency. Trading can aggregate dispersed information and generate informative forecasts, while observed prices may still reflect calibration biases, heterogeneous beliefs, liquidity conditions, transaction costs, and market design. These qualifications become especially relevant when prediction market valuations are compared with a benchmark constructed from another financial market.

3.3 Modern Prediction Markets: Evidence from Polymarket and Kalshi

The expansion of Polymarket and Kalshi has made it possible to study prediction markets at a level of detail that was generally unavailable in the earlier literature. Recent work uses transaction-level and order-book data to examine not only forecast accuracy, but also the role of liquidity, trading behaviour, and market microstructure in price formation. This evidence is particularly useful because informative aggregate prices can coexist with local pricing distortions and substantial differences across traders.

Using Kalshi contracts linked to macroeconomic releases, Diercks et al. (2026) compare prediction-market forecasts with survey expectations and other market-based measures. They find that Kalshi provides high-frequency, continuously updated, and distributionally rich information about macroeconomic expectations. Rather than replacing conventional forecasts, these contracts offer an additional measure that can be observed and updated between scheduled surveys and data releases.

A broader analysis of Kalshi is provided by Bürgi et al. (2026), who study transaction-level data covering more than 300,000 contracts. Prices are informative and become more accurate as markets approach resolution, but they also display a clear favourite–longshot bias. Low-priced contracts win less frequently than their prices would imply, whereas high-priced contracts perform relatively better. The authors relate these patterns to Kalshi's maker–taker microstructure, in which makers post offers and takers choose whether to trade against them. Their results show that average forecasting accuracy does not rule out systematic differences across price ranges or trading roles.

Recent research on Polymarket examines similar questions using its hybrid-decentralised architecture, which combines an off-chain central limit order book with on-chain settlement. Based on transaction data from the 2024 U.S. presidential election, Tsang and Yang (2026) distinguish exchange-equivalent turnover from the minting and burning of contract shares. This distinction matters because a simple aggregation of blockchain transactions can overstate economically meaningful trading activity. Over the election period, the half-lives of internal arbitrage deviations fell from several hours to less than one minute, while Kyle's λ declined substantially. Internal pricing consistency and market

depth therefore improved as trading activity developed.

Akey et al. (2026) examine the distribution of trading gains and losses on Polymarket using approximately $67 billion in trading volume. The gains are highly concentrated: the top 1% of users capture 76.5% of total profits. Successful traders tend to provide liquidity through limit orders, whereas unsuccessful traders are more likely to take liquidity using market orders. Monthly performance is only modestly persistent, and the authors caution that part of this persistence may reflect selection rather than skill. Their findings reveal substantial heterogeneity behind the prices observed at the platform level.

A different aspect of price discovery is studied by Aktuğ and Torul (2026). The authors examine the inadvertent early release of the May 2024 U.S. CPI report. CME Federal Funds futures responded within seconds of the information becoming available, while the first detectable adjustment in the corresponding Polymarket contracts occurred at 08:35 ET. The evidence is consistent with limited monitoring and updating in a thin retail prediction market. However, the study concerns a single episode and does not support a general conclusion about Polymarket's informational efficiency.

The recent evidence therefore does not support a simple classification of either platform as efficient or inefficient. Kalshi can provide timely and informative macroeconomic expectations while still displaying systematic pricing patterns across probability levels. Polymarket's market quality appears to improve with liquidity, but trading outcomes remain highly concentrated and information incorporation may be slow in particular episodes. Forecast quality and price formation depend on liquidity, contract design, participation, and the conditions under which information reaches the market.

These findings support the use of modern prediction-market prices as informative market valuations, while leaving room for economically meaningful deviations across contracts and platforms. The next subsection turns to derivatives-implied information and the construction of an external financial-market benchmark.

3.4 Derivatives-Implied Information and Cryptocurrency Options

Option prices provide a market-based source of forward-looking information about the distribution of an underlying asset. Unlike historical volatility estimates, which are constructed from past returns, option prices reflect the terms on which market participants are willing to trade claims on future states. The information they contain is therefore forward-looking, although it should not be interpreted as a direct forecast of the physical distribution of future returns.

The theoretical link between option prices and state-contingent valuations is established by Breeden and Litzenberger (1978). Under standard no-arbitrage conditions, a suffi-

ciently rich cross-section of European option prices with a common maturity can be used to recover state prices and, up to discounting, the risk-neutral density assigned to the future value of the underlying asset. Digital option prices are closely related to this result because they represent claims on specific terminal events rather than continuous payoffs.

Recovering an option-implied distribution in practice is less straightforward. Observed option prices are available only for a limited set of strikes and maturities, while market quotes may be affected by bid–ask spreads, sparse trading, and measurement noise. Estimation therefore requires interpolation or smoothing across the option surface. Aït-Sahalia and Lo (1998), for example, develop a non-parametric approach to estimating state-price densities from option prices, allowing the implied distribution to depart from the lognormal benchmark.

A separate issue concerns the meaning of the recovered distribution. Option prices identify risk-neutral state valuations rather than physical event probabilities. Risk-neutral probabilities incorporate the pricing of risk and need not coincide with investors' actual beliefs about future outcomes. Moving from one distribution to the other requires assumptions about the pricing kernel or investors' risk preferences. The analysis of Bliss and Panigirtzoglou (2004) illustrates how estimates of risk aversion can be used to adjust option-implied distributions, but such a conversion is necessarily model-dependent.

This distinction is particularly relevant when derivatives-implied probabilities are compared with prediction market prices. A difference between the two does not, by itself, establish that either market is incorrect. It may reflect risk premia in option prices, different investor populations, liquidity conditions, or differences in the information incorporated by each venue. The comparison remains useful, but its interpretation must recognise that the two prices need not represent the same probability concept.

Cryptocurrency option markets make this type of analysis possible for Bitcoin. Deribit supports an active market for Bitcoin options and publishes the Bitcoin DVOL index, a rolling 30-day annualised measure of implied volatility derived from option prices (Fernando 2021). DVOL summarises forward-looking volatility information across part of the option market in a single observable series.

DVOL is not a complete Bitcoin option-implied distribution. In particular, it does not preserve the full strike and maturity structure of the option surface, nor does it separately describe skewness and tail probabilities. It is an aggregate volatility measure. A probability associated with a specific threshold cannot therefore be read directly from the index.

For the purposes of this thesis, DVOL is not treated as a direct estimate of the probability associated with a specific Bitcoin price threshold. It is combined with the latest available

Bitcoin spot price and the remaining time to maturity in a lognormal terminal-price model. The resulting measure is referred to as a DVOL-based benchmark probability rather than as a probability directly implied by a particular Deribit option.

This parsimonious construction provides a common and replicable benchmark for a large set of prediction market contracts. It does not, however, retain strike-specific implied volatility, volatility skew, or the full maturity structure of the Bitcoin option market. The benchmark is therefore interpreted as an external derivatives-based reference, not as a model-free estimate of the physical probability or as a complete risk-neutral distribution.

3.5 Cross-Market Pricing in Cryptocurrency Prediction Markets

A small but growing literature compares cryptocurrency prediction market prices with valuations obtained from derivatives markets. This setting is particularly useful because the event contracts are written on observable financial assets for which active spot and derivatives markets also exist. Unlike comparisons between prediction markets and surveys, both sides of the comparison are therefore generated through market trading.

The two valuations are not expected to coincide mechanically. Prediction market prices and option-implied quantities are formed under different pricing environments and may reflect different probability concepts, market participants, and trading frictions. Cross-market differences must therefore be interpreted more cautiously than violations of a simple law-of-one-price relationship.

The study most closely related to this thesis is Fabi et al. (2026). The authors compare Polymarket prices with benchmarks obtained from option-implied risk-neutral distributions for cryptocurrency price events. In the current version of the paper, the analysis covers contracts written on Bitcoin and Ethereum. Polymarket prices generally move closely with the option-implied benchmarks, but systematic differences remain. These are more pronounced for tail and barrier events and during periods of high volatility or major macroeconomic shocks, and are associated with sentiment, attention, demand, and market-specific frictions.

Payoff structure is an important part of their analysis. Terminal price contracts can be compared with probabilities obtained from a terminal distribution, whereas contracts that depend on whether a threshold is reached before expiration require a path-dependent valuation. Fabi et al. (2026) distinguish explicitly between terminal *Above* and *Range* contracts and path-dependent *Reach* and *Dip* contracts. Although average cross-market correlations remain high across all four categories, the correlation distributions for *Reach* and *Dip* contracts are more dispersed and include more intermediate, low, and occasionally negative values. Payoff structure is therefore relevant when interpreting differences between Polymarket prices and the option-implied benchmark.

Valery (2025) approaches cross-market pricing from a different direction. The paper studies pricing discrepancies between cryptocurrency inverse vanilla options and binary options traded on prediction markets. It derives a condition under which the two instruments can be combined into a portfolio with a non-negative terminal payoff and then evaluates a trading strategy based on that condition using Bitcoin and Ether contracts. In the author's empirical tests, such opportunities occur relatively infrequently, although none of the executed trades produces a negative terminal return.

The difference between these two approaches is economically important. Fabi et al. (2026) use the derivatives market primarily to construct option-implied benchmarks against which prediction market prices can be evaluated. Valery (2025), by contrast, focuses on payoff relations between vanilla and binary options and on the trading opportunities that may arise when those relations are violated. Both study cross-market consistency, but a measured difference between probability-like valuations is not equivalent to an executable arbitrage strategy once liquidity, bid–ask spreads, transaction costs, and execution constraints are taken into account.

The present thesis belongs to the probability-comparison strand of this literature rather than to the construction of an arbitrage portfolio. Its specific empirical design and expectations are set out in the next subsection.

3.6 Positioning of the Thesis and Empirical Expectations

The literature reviewed in this chapter provides two main points of reference for the present analysis. Research on prediction markets shows that contract prices can aggregate dispersed information and produce informative forecasts, although biases, liquidity constraints, and institutional differences may still affect observed prices. The option-pricing literature offers a separate source of forward-looking information, but option-implied quantities are risk-neutral and depend on the characteristics of the available derivatives data.

The thesis is most closely related to the probability-comparison approach developed by Fabi et al. (2026). It differs from that study by using a more parsimonious benchmark based on Bitcoin spot prices, Deribit DVOL, and a lognormal terminal-price model, rather than recovering a complete distribution from the historical option surface. It also differs from Valery (2025), since the analysis measures differences between probability-like valuations instead of constructing and testing an arbitrage portfolio.

Signed mispricing is defined as the prediction market probability minus the corresponding DVOL-based benchmark probability. A non-zero value records a difference between two market-based valuations; it does not establish the existence of an executable arbitrage opportunity. Transaction costs, liquidity, execution conditions, and differences in contract

specifications remain relevant to its interpretation.

Payoff comparability determines the organisation of the empirical samples. Polymarket terminal contracts form the main sample because their payoffs can be represented using a terminal price distribution. Path-dependent contracts are retained separately, since applying the same benchmark creates a mismatch between their payoff structure and the probability model. Kalshi provides a supplementary regulated-market comparison, but its short trading horizons, bin-based contracts, and different market design make it imperfectly comparable with the main Polymarket sample.

All cross-market comparisons use benchmark information available at the relevant observation time, avoiding the use of future information. The operational implementation is described in the Data and Methodology chapters.

Empirical expectations. Prediction market prices and the benchmark probabilities need not coincide exactly. Some degree of co-movement may be expected because both valuations refer to economically related Bitcoin price events, while differences may remain because the markets involve different participants, trading conditions, risk exposures, and information sets. Path-dependent contracts may display larger deviations, but any such differences must be interpreted in light of the mismatch between their payoff structure and a terminal benchmark. Mispricing may also vary across Polymarket and Kalshi because the two platforms operate under different institutional arrangements. The regression analysis is exploratory and describes conditional associations rather than causal effects.

The thesis therefore examines cross-market pricing consistency using a transparent benchmark while keeping payoff structure and institutional comparability explicit. The following chapters describe the data, benchmark construction, and empirical implementation.

4 Data and Sample Construction

This section describes the data sources, the construction of the empirical samples, and the temporal alignment used to compare Bitcoin prediction market prices with a DVOL-based derivatives benchmark. The dataset combines information from Polymarket, Kalshi, Binance, and Deribit. Polymarket and Kalshi provide observed prices for Bitcoin event contracts, Binance supplies the underlying Bitcoin spot prices, and Deribit's DVOL index provides the implied-volatility input used to calculate the benchmark probabilities.

The contracts differ in their wording, payoff structure, and trading horizon. Each observation is therefore reduced to a common set of elements: an observable Bitcoin price, an identified threshold or interval, and a defined expiration horizon. Temporal consistency is essential. A benchmark input can be assigned to a prediction market observation only if it was already available when that market price was recorded.

The sample construction also separates terminal contracts from path-dependent contracts. A terminal claim depends only on the Bitcoin price at expiration and can therefore be compared with a probability calculated from a terminal lognormal distribution. A path-dependent claim instead depends on whether Bitcoin reaches a threshold at any point before expiration. Because the terminal benchmark does not reproduce this payoff, these contracts are retained in a separate comparison sample.

4.1 Data Sources

The empirical dataset draws on four sources. Polymarket provides contract metadata and historical prices for Bitcoin-related events. These records contain the contract description, observed market price, expiration date, and the information needed to identify the relevant strike threshold or interval.

Kalshi supplies contract metadata and timestamped historical transactions for daily Bitcoin price-range events in the KXBTC series. The trades are collected through Kalshi's official historical trades endpoint (Kalshi 2026). Kalshi is included as a supplementary comparison from a prediction market operating within a regulated institutional framework.

Bitcoin spot prices are obtained through the Kline/Candlestick endpoint of the Binance Spot REST API (Binance 2026). The required frequency differs between the two prediction market samples. Daily observations are used for Polymarket, while one-minute prices allow the Kalshi benchmark to be aligned with the timestamp of each selected transaction.

Deribit's DVOL index is the final data source. It provides an aggregate measure of Bitcoin implied volatility. The analysis does not reconstruct a complete historical option-

implied volatility surface. DVOL is instead used as a transparent volatility proxy that is consistently available over the sample period.

Because DVOL does not provide implied volatilities specific to every strike and maturity, the resulting probabilities are interpreted as a DVOL-based derivatives benchmark. They should not be regarded as a complete set of risk-neutral probabilities recovered from Deribit's historical option surface.

4.2 Polymarket Data

Polymarket provides the principal prediction market sample. The analysis includes contracts whose payoff is linked to the future price of Bitcoin. Relevant markets are identified from contract metadata and historical price records, while the contract descriptions are parsed to recover the payoff type, strike threshold or interval, and expiration date.

For each observation, the Polymarket probability is measured by the historical price recorded for the contract. A single contract can be observed at several timestamps, so the resulting data have a panel structure. Only prices within the economically valid interval from zero to one are retained.

Contracts are divided according to whether their payoff is terminal or path-dependent. The terminal sample contains *above* and *range* markets. Since these contracts depend only on the Bitcoin price at expiration, they provide the closest comparison with the terminal probability benchmark.

The path-dependent sample consists of *reach* and *dip* markets. Their payoff depends on Bitcoin touching a specified threshold before the contract expires. These observations remain informative as a separate comparison, but their payoff cannot be represented exactly by the terminal benchmark.

The raw Polymarket analysis dataset contains 30,060 price observations. The temporal filter removes 88 observations with zero or negative remaining time to maturity. The matched dataset consequently contains 29,972 observations from 3,431 markets. Of these, 19,649 observations from 2,847 markets belong to the terminal sample. The path-dependent sample contains the remaining 10,323 observations from 584 markets.

4.3 Kalshi Data

Kalshi is included as a supplementary comparison from a regulated prediction market. The sample covers daily Bitcoin price events in the KXBTC series. Most contracts refer to mutually exclusive terminal price intervals, with lower- and upper-tail contracts included when available.

The sample is constructed from timestamped historical transactions rather than from current metadata fields or retrospectively reported quotes. For each contract with valid trading activity, the pipeline identifies the first valid trade after the market opens. The corresponding Yes price is used as the observed Kalshi probability.

The baseline sample requires the first valid trade to occur within 30 minutes of market opening. This restriction leaves 3,135 contracts from 485 daily events. The median selected transaction occurs approximately 4.6 minutes after opening, and all retained trades satisfy the 30-minute limit. Since the markets remain open for one hour, these observations have between approximately 30 and 60 minutes remaining until close.

The transaction records are checked before entering the sample. Timestamps and trade identifiers must be valid, prices and quantities must lie within their admissible ranges, and the downloaded transactions must be consistent with the relevant contract metadata. Contracts are removed if they fail the required reconciliation checks or do not record a valid trade within the selected window.

A narrower 15-minute window is used as a timing robustness check. It contains 2,554 contracts from 483 events and tests whether the baseline result is sensitive to transactions occurring later in the one-hour trading period.

The Kalshi sample is not directly equivalent to the Polymarket terminal sample. Kalshi contracts have much shorter horizons, follow a different bin structure, and trade within another institutional framework. Their results are therefore treated as supplementary evidence rather than as the main finding of the thesis.

4.4 Bitcoin Spot Prices and Deribit DVOL

Calculating the derivatives-based benchmark requires a Bitcoin spot price and a measure of expected volatility. Spot prices are obtained from Binance, while expected volatility is proxied by the Deribit Bitcoin Volatility Index. Since DVOL is reported as an annualised percentage, it is converted into decimal form:

$$\sigma_t = \frac{\mathrm{DVOL}_t}{100}. \tag{10}$$

DVOL is an aggregate implied-volatility index rather than a volatility surface specific to each strike and maturity. The resulting benchmark is therefore an approximation based on information from the Bitcoin derivatives market. It should not be interpreted as a complete probability distribution recovered from historical option prices.

The timing convention differs between Polymarket and Kalshi. Each Polymarket observation is assigned the latest daily Bitcoin spot price and DVOL value that had already

become available by its timestamp. Matching is based on information availability, preventing an observation from being paired with information from a trading day that had not yet been completed.

Kalshi transactions have exact timestamps. The spot input is the latest one-minute Bitcoin price recorded at or before the selected trade. Since DVOL is available only at the daily frequency, each transaction is assigned the most recent daily value already observable at that time.

Every benchmark input must satisfy

$$t_j^{\text{available}} \leq t_i^{\text{observation}}, \tag{11}$$

where $t_i^{\text{observation}}$ denotes the time of prediction market observation i, and $t_j^{\text{available}}$ is the time at which benchmark input j became available. This restriction prevents the benchmark from incorporating future information.

No spot or volatility inputs are missing from the final Kalshi benchmark. The one-minute spot observation is at most 60 seconds older than the corresponding Kalshi trade, while the matched DVOL observation is between roughly three and four and a half hours old. The timing rule prevents look-ahead bias, although the latest available daily DVOL value may be somewhat stale relative to the intraday transaction.

4.5 Contract Classification

Contracts must be classified before an appropriate benchmark probability can be assigned. Although all observations refer to Bitcoin prices, their payoff structures differ and cannot always be represented using the same terminal probability formula.

A Polymarket *above* contract pays if Bitcoin finishes above a specified threshold at expiration. It is a terminal claim and is comparable to a digital call option. A *range* contract pays when the expiration price lies between two boundaries, so its benchmark is based on the probability mass between a lower and an upper strike.

The *reach* and *dip* categories have a different payoff structure. A *reach* contract depends on Bitcoin touching an upper threshold before expiration, whereas a *dip* contract depends on its price falling below a lower threshold during the life of the contract. Both are path-dependent and are more closely related to barrier or touch options than to terminal digital claims.

Kalshi contracts are classified as *greater*, *less*, or *between*. The first two represent upper- and lower-tail terminal events using a single strike boundary. A *between* contract defines a terminal interval through a floor strike and a cap strike.

The principal empirical sample contains Polymarket *above* and *range* contracts. The *reach* and *dip* observations are analysed separately as a path-dependent comparison. Kalshi provides an additional short-horizon sample from a regulated market.

4.6 Matching Procedure

The matching procedure combines each prediction market observation with its contract characteristics and the inputs required to calculate the DVOL-based benchmark probability. These elements include the relevant Bitcoin threshold or interval, the observation and expiration times, the latest available spot price, and the applicable DVOL value.

A Polymarket *above* contract is represented by a single threshold K. A *range* contract instead requires a lower boundary K_L and an upper boundary K_U. The *reach* and *dip* categories also contain a single threshold, but they are identified as path-dependent because the timing of a threshold crossing affects their payoff.

The same distinction is applied to Kalshi. *Greater* and *less* contracts use one relevant strike boundary, while *between* contracts use a floor strike and a cap strike. Their observation time is the timestamp of the first valid trade recorded within 30 minutes of market opening.

Time to maturity is measured from the exact observation and expiration timestamps:

$$\tau_i = \frac{t_i^{\text{expiry}} - t_i^{\text{observation}}}{365 \times 24 \times 60 \times 60}. \tag{12}$$

Observations with zero or negative remaining time to maturity are excluded. For Polymarket, the expiration time is obtained from the contract end time. The Kalshi probability calculation instead uses the market closing time as the expiration time.

Benchmark inputs are assigned according to when they became available. Polymarket observations use the latest available daily spot and DVOL values. Kalshi trades use the latest one-minute spot observation and the latest available daily DVOL value at the transaction timestamp.

After matching, each retained observation contains the prediction market probability, observation time, strike threshold or interval, time to maturity, available Bitcoin spot price, DVOL-based volatility input, and corresponding derivatives-implied benchmark probability.

4.7 Sample Filters and Exclusions

The final samples are subject to filters designed to preserve economic meaning, payoff comparability, and temporal validity.

Only Bitcoin-related contracts with an identifiable payoff structure are retained. Each observation must have a valid threshold or interval, an expiration time, and an observed prediction market probability. Prices must lie between zero and one, and the strike boundaries must be economically admissible.

Time to maturity is required to be strictly positive. This condition removes 88 Polymarket observations recorded at or after the relevant expiration time.

Benchmark variables are also subject to an information-availability condition. Spot prices and DVOL values must have become available no later than the corresponding prediction market observation. Records with missing inputs or violations of this timing rule are excluded. No such violations remain in the final matched datasets.

Polymarket observations are then separated by payoff structure. Terminal contracts form the main sample, while path-dependent contracts are retained only for the separate comparison.

A Kalshi contract must have a valid historical transaction that passes the reconciliation checks. The 30-minute baseline retains contracts whose first valid trade occurs no later than 30 minutes after market opening. Contracts without a qualifying transaction are removed. The 15-minute robustness sample follows the same rules with a narrower timing window.

These restrictions ensure that the analysis uses observed transaction or historical market prices rather than retrospective metadata. They also guarantee that the benchmark inputs were available at the relevant observation time.

4.8 Final Empirical Samples

Table 1 summarises the datasets used in the empirical analysis.

Sample	Observations	Distinct contracts or markets
Polymarket terminal contracts	19,649	2,847
Polymarket path-dependent contracts	10,323	584
Kalshi first trade within 30 minutes	3,135	3,135
Kalshi first trade within 15 minutes	2,554	2,554
Unified baseline dataset	33,107	6,566

Table 1: Final empirical samples. The Kalshi 15-minute sample is a robustness subset of the 30-minute baseline sample.

The Polymarket terminal dataset is the main empirical sample because its payoff structure can be compared directly with the terminal benchmark. The path-dependent sample serves a different purpose, showing how measured deviations change when the same benchmark is applied to claims whose payoff depends on the price path.

The Kalshi baseline contains 3,135 traded contracts from 485 daily events. Contracts belonging to the same event are related, so the statistical analysis accounts for dependence at the event level. The narrower 15-minute sample contains 2,554 contracts from 483 events.

Combining the Polymarket observations with the Kalshi baseline produces a unified dataset containing 33,107 observations and 6,566 distinct contracts or markets. These observations correspond to 3,916 empirical clusters: 3,431 Polymarket markets and 485 Kalshi events. The unified dataset is used for cross-market comparisons, although the main interpretation continues to rely on Polymarket terminal contracts.

4.9 Data Limitations

The dataset has several limitations. Polymarket markets contribute different numbers of historical observations, so repeated prices from the same market cannot be treated as independent. The statistical analysis addresses this dependence through inference conducted at the market level.

The Kalshi sample includes only contracts that record a valid historical trade within 30 minutes of opening. Although this requirement provides a clearly timed market price, it may favour contracts with greater trading activity. The first observed trade may also contain idiosyncratic noise and does not necessarily represent a frictionless consensus valuation. The 15-minute sample provides a check on the sensitivity of the results to the selected timing window.

There are also substantial differences between the two prediction markets. Kalshi contracts have shorter horizons, are organised into mutually exclusive price bins, and trade

under another institutional structure. Their results are therefore used as supplementary evidence rather than as a substitute for the Polymarket terminal analysis.

The benchmark is necessarily approximate. DVOL is an aggregate measure of implied volatility and does not reproduce strike-specific volatility, skew, or the complete maturity structure of the Bitcoin options market. It should not be interpreted as a full risk-neutral density recovered from historical option prices.

The daily frequency of DVOL introduces an additional timing limitation, particularly for the intraday Kalshi observations. The latest available value may be several hours old when a selected trade occurs. The information-availability restriction prevents future data from entering the benchmark, but it cannot make the volatility input exactly contemporaneous with each transaction.

The terminal benchmark is also not designed for Polymarket *reach* and *dip* contracts. Their payoffs depend on whether a threshold is crossed before expiration, not only on the terminal Bitcoin price. Results for these contracts are reported separately and do not form part of the primary evidence.

The analysis does not fully model bid–ask spreads, market depth, transaction costs, execution constraints, or the practical replication of identical payoffs across venues. Measured mispricing should therefore be interpreted as a difference between probability-like market valuations, not as a directly executable arbitrage return.

5 Methodology

This section explains how Bitcoin prediction market prices are compared with a DVOL-based derivatives benchmark. For every matched observation, the calculation combines the observed prediction market price with the latest available Bitcoin spot price, Deribit DVOL, the relevant strike boundary or interval, and the remaining time to maturity.

Benchmark probabilities are calculated under a terminal lognormal framework, with DVOL used as the annualised volatility input. The procedure does not reconstruct a complete risk-neutral density from historical Deribit option prices. It instead produces a transparent probability proxy based on derivatives-market information. The proxy should not be interpreted as an exact probability extracted from the full Deribit volatility surface.

5.1 Observed Prediction Market Probability

For contract i observed at time t, let the observed prediction market probability be

$$P_{i,t}^{M}. \tag{13}$$

The contracts considered in the thesis pay one unit if the specified event occurs and zero otherwise. Subject to the conditions discussed in Section 2.2, their prices can therefore be treated as probability-like market valuations.

For Polymarket, $P_{i,t}^{M}$ is the historical market price recorded at the corresponding observation timestamp. A single market may contribute several observations because its price can be recorded at different times. All retained prices lie between zero and one.

The Kalshi probability is defined differently. For each contract, $P_{i,t}^{M}$ is the Yes price of the first valid timestamped historical trade after the market opens. The 30-minute baseline sample imposes the following timing condition:

$$0 < t_i^{\text{trade}} - t_i^{\text{open}} \leq 30 \text{ minutes}. \tag{14}$$

Contracts without a valid transaction during this window are excluded. A narrower 15-minute window is used for the timing robustness check. The Kalshi probability is not based on a retrospectively reported quote, a current bid–ask midpoint, or an updated metadata price.

Observed prices need not be unbiased estimates of physical event probabilities. Risk preferences, liquidity, transaction costs, market segmentation, behavioural demand, and venue-specific trading constraints may all affect them. The empirical analysis con-

sequently treats $P_{i,t}^{M}$ as an observed probability-like valuation rather than as a direct measure of the real-world probability of the event.

5.2 DVOL-Based Benchmark Probability

Let S_t be the latest Bitcoin spot price available at time t, K the relevant strike threshold, σ_t the annualised volatility proxy obtained from DVOL, and $\tau_{i,t}$ the remaining time to maturity in years.

DVOL is reported in percentage terms and is converted into decimal form:

$$\sigma_t = \frac{\mathrm{DVOL}_t}{100}. \tag{15}$$

For a terminal event in which $S_T > K$, the benchmark probability is

$$P_{i,t}^{\mathrm{Deribit}}(S_T > K) = \Phi(d_{2,i,t}), \tag{16}$$

where

$$d_{2,i,t} = \frac{\ln(S_t/K) + \left(r - \frac{1}{2}\sigma_t^2\right)\tau_{i,t}}{\sigma_t\sqrt{\tau_{i,t}}}, \tag{17}$$

and $\Phi(\cdot)$ denotes the cumulative distribution function of the standard normal distribution.

The risk-free rate is set to zero:

$$r = 0. \tag{18}$$

Given the predominantly short horizons in the sample, this assumption has limited numerical relevance and keeps the construction of the benchmark transparent. Approximation remains unavoidable, however, because DVOL does not account for strike-specific implied volatility, volatility skew, or maturity-specific variation in Bitcoin option prices.

Throughout the analysis, $P_{i,t}^{\mathrm{Deribit}}$ is interpreted as a DVOL-based, risk-neutral-style probability proxy. It is not treated as a physical event probability or as a complete probability recovered from the historical Deribit option chain.

5.3 Tail and Range Probabilities

For a lower-tail terminal event of the form $S_T < K$, the benchmark probability is

$$P_{i,t}^{\text{Deribit}}(S_T < K) = 1 - \Phi(d_{2,i,t}) = \Phi(-d_{2,i,t}). \tag{19}$$

The lower-tail expression is used for the relevant Polymarket terminal contracts and for Kalshi *less* contracts.

An interval event is defined by

$$K_L < S_T < K_U. \tag{20}$$

Its benchmark probability is the difference between the cumulative probabilities evaluated at the upper and lower boundaries:

$$P_{i,t}^{\text{Deribit}}(K_L < S_T < K_U) = P_{i,t}^{\text{Deribit}}(S_T < K_U) - P_{i,t}^{\text{Deribit}}(S_T < K_L). \tag{21}$$

Equivalently,

$$P_{i,t}^{\text{Deribit}}(K_L < S_T < K_U) = \Phi(-d_2(K_U)) - \Phi(-d_2(K_L)). \tag{22}$$

This expression is applied to Polymarket *range* contracts and Kalshi *between* contracts. Kalshi *greater* and *less* contracts are evaluated with the corresponding upper- and lower-tail formulas.

Because the terminal distribution is continuous, the probability assigned exactly to a strike boundary is zero. Strict and weak inequalities therefore produce the same calculation.

5.4 Time to Maturity and Information Availability

Time to maturity is calculated from the exact observation and expiration timestamps:

$$\tau_{i,t} = \frac{t_i^{\text{expiry}} - t_i^{\text{observation}}}{365 \times 24 \times 60 \times 60}. \tag{23}$$

Observations with $\tau_{i,t} \leq 0$ are removed. The Polymarket expiration timestamp comes from the contract end time, while the Kalshi probability calculation uses the market closing time.

Only information already available at the prediction market timestamp enters the benchmark. Each input j must satisfy

$$t_j^{\text{available}} \leq t_i^{\text{observation}}. \tag{24}$$

Polymarket observations use the latest daily spot and DVOL values that had become available by the corresponding timestamp. For Kalshi, S_t is the latest one-minute Bitcoin spot observation recorded at or before the selected trade. The volatility input σ_t comes from the latest available daily DVOL value.

This restriction prevents future information from entering the benchmark. It also creates a trade-off: the daily DVOL input may be somewhat stale, particularly for the intraday Kalshi trades.

5.5 Treatment of Path-Dependent Contracts

Polymarket *reach* and *dip* contracts are separated from the terminal sample because their payoff depends on the Bitcoin price path before expiration. A *reach* contract pays when Bitcoin touches an upper threshold during the life of the contract; a *dip* contract pays when the price touches a lower threshold.

Such payoffs are closer to barrier or touch options than to terminal digital options. A terminal probability calculated from the expiration price is therefore not theoretically tailored to these contracts.

The empirical pipeline nevertheless assigns *reach* and *dip* observations the corresponding terminal upper- or lower-tail probability. This calculation is used only as a comparison exercise. Any resulting deviation must therefore be interpreted in light of the mismatch between the payoff structure and the benchmark. It cannot be read as evidence that the terminal proxy incorrectly prices an equivalent claim. The principal empirical inference is based on the Polymarket terminal sample.

5.6 Mispricing Measure

The main variable of interest is the signed difference between the observed prediction market probability and the DVOL-based benchmark:

$$\text{Mispricing}_{i,t} = P_{i,t}^{M} - P_{i,t}^{\text{Deribit}}. \tag{25}$$

Positive mispricing means that the prediction market assigns a higher probability to the event than the benchmark. A negative value means that the observed prediction market probability is lower.

The absolute deviation is

$$\left|\text{Mispricing}_{i,t}\right| = \left|P_{i,t}^{M} - P_{i,t}^{\text{Deribit}}\right|. \tag{26}$$

Signed mispricing preserves the direction of the difference, whereas absolute mispricing measures its size without regard to direction.

Neither variable represents a directly executable arbitrage profit. Contracts may not be perfectly replicable across venues, and the calculations do not fully account for bid–ask spreads, fees, market depth, collateral requirements, execution risk, short-selling constraints, or differences in settlement.

5.7 Statistical Inference

The descriptive analysis reports observation-level means, medians, standard deviations, distributions, and changes over time. Conventional observation-level tests and row-level bootstrap confidence intervals are included as baseline diagnostics.

Those procedures may overstate precision when applied in isolation. Polymarket contributes repeated price observations from the same markets, while Kalshi contracts associated with a single daily event represent mutually related outcomes. Preferred inference is therefore conducted at the cluster level.

For Polymarket market c, average mispricing is defined as

$$\overline{M}_c = \frac{1}{N_c} \sum_{(i,t)\in c} \text{Mispricing}_{i,t}. \tag{27}$$

The principal Polymarket estimate gives equal weight to each market by averaging these market-level means:

$$\widehat{\mu}_P = \frac{1}{C_P} \sum_{c=1}^{C_P} \overline{M}_c. \tag{28}$$

Confidence intervals are calculated by resampling Polymarket market identifiers with replacement. Markets with more frequent observations therefore do not automatically receive greater inferential weight.

For Kalshi, dependence occurs at the daily-event level rather than at the level of the individual ticker. Contracts within the same event correspond to mutually exclusive price bins and cannot reasonably be treated as independent observations. Mispricing is first averaged within each event and then across events:

$$\widehat{\mu}_K = \frac{1}{C_K} \sum_{c=1}^{C_K} \overline{M}_c. \tag{29}$$

Event identifiers are resampled with replacement to construct the Kalshi confidence intervals.

The unified sample uses a hybrid inference unit. Polymarket observations are clustered by market identifier, whereas Kalshi observations are clustered by event identifier. This definition follows the different dependence structures of the two datasets.

5.8 Regression Analysis

The regressions examine conditional associations between signed mispricing and benchmark probability, time to maturity, implied volatility, contract type, and market environment.

For the Polymarket terminal sample, the main specification is

$$\text{Mispricing}_{i,t} = \alpha + \beta_1 P_{i,t}^{\text{Deribit}} + \beta_2 \text{TauDays}_{i,t} + \beta_3 \sigma_t + \gamma \text{Range}_i + \varepsilon_{i,t}, \tag{30}$$

where Range_i equals one for Polymarket *range* contracts and zero for *above* contracts. Standard errors are clustered by Polymarket market identifier.

The unified specification is

$$\begin{aligned} \text{Mispricing}_{i,t} = \alpha &+ \delta_1 \text{Polymarket}_i + \delta_2 \text{PathDependent}_i \\ &+ \beta_1 P_{i,t}^{\text{Deribit}} + \beta_2 \text{TauDays}_{i,t} + \beta_3 \sigma_t + \varepsilon_{i,t}. \end{aligned} \tag{31}$$

Here, δ_1 measures the conditional difference between Polymarket terminal observations and Kalshi observations, with Kalshi as the reference category. The coefficient δ_2 captures the additional difference associated with Polymarket path-dependent contracts relative to Polymarket terminal contracts.

Because $P_{i,t}^{\text{Deribit}}$ also enters the definition of the dependent variable, the coefficient β_1 should be interpreted with particular caution. Part of its estimated association with signed mispricing is mechanical. It is therefore not treated as an independent structural determinant of mispricing.

Standard errors in the unified regression follow the hybrid clustering rule: Polymarket market identifiers are used for Polymarket observations, while Kalshi event identifiers are

used for Kalshi observations. Specifications based on heteroskedasticity-robust covariance estimators are reported in Appendix A.

These regressions are descriptive, not causal. Their coefficients summarise conditional associations and cross-market differences within the constructed samples; they do not identify structural determinants of prediction market prices.

5.9 Robustness Checks

The empirical findings are evaluated under several alternative sample and inference choices.

The principal analysis uses only Polymarket terminal contracts, with *reach* and *dip* observations examined separately. A second Polymarket sample restricts the terminal observations to those with no more than seven days remaining until expiration.

Inference is also adjusted for the dependence structure of each market. Polymarket confidence intervals are obtained through market-level bootstrap resampling, while Kalshi confidence intervals use event-level resampling. The monthly Polymarket analysis provides a more aggregated check. In its filtered version, only calendar months containing at least 100 observations are retained, reducing the influence of sparsely represented periods.

For Kalshi, the baseline uses the first valid transaction recorded within 30 minutes of market opening. Repeating the analysis with a 15-minute window tests whether the result is sensitive to trades selected later in the one-hour session.

Regression estimates are reported with both heteroskedasticity-robust and cluster-robust covariance estimators. The clustered specifications are preferred because they account for repeated Polymarket observations and dependence among Kalshi contracts belonging to the same event.

6 Empirical Results

This section compares Bitcoin prediction market probabilities with the DVOL-based derivatives benchmark developed in the previous chapter. The main analysis concerns Polymarket terminal contracts, whose payoffs correspond most closely to the terminal probabilities calculated by the benchmark. Kalshi provides a supplementary comparison from a regulated prediction market. Polymarket path-dependent contracts are examined separately because their payoff structure is not captured by a terminal model.

The analysis considers both statistical precision and economic magnitude. Particular attention is given to the dependence created by repeated Polymarket observations and by Kalshi contracts belonging to the same daily event.

6.1 Overview of Empirical Samples

Table 2 reports the number of observations and the preferred inference clusters for each empirical sample. Polymarket observations are clustered by market, whereas Kalshi contracts are clustered by daily event because contracts belonging to the same event represent mutually related price bins. The Kalshi 15-minute sample is a subset of the 30-minute baseline and is used only as a timing robustness check; it is therefore not added separately to the unified baseline dataset.

Sample	**Observations**	**Inference clusters**
Polymarket terminal	19,649	2,847
Polymarket terminal, $\tau \leq 7$ days	19,616	2,847
Polymarket path-dependent	10,323	584
Kalshi, first trade within 30 minutes	3,135	485
Kalshi, first trade within 15 minutes	2,554	483
Unified baseline dataset	33,107	3,916

Table 2: Overview of the empirical samples and preferred inference clusters. Polymarket observations are clustered by market identifier, while Kalshi observations are clustered by daily event identifier.

The unified baseline combines 29,972 Polymarket observations with 3,135 Kalshi observations. Its 3,916 inference clusters consist of 3,431 Polymarket markets and 485 Kalshi daily events. This clustering structure accounts for repeated observations from the same Polymarket market and for dependence among Kalshi contracts belonging to the same event.

6.2 Descriptive Evidence on Mispricing

Table 3 reports observation-level descriptive statistics for the Polymarket terminal sample and the Kalshi 30-minute baseline. These statistics describe the underlying observations, whereas the preferred confidence intervals reported later are constructed at the Polymarket-market or Kalshi-event level.

Variable	**Mean**	**Median**	**SD**	**P25**	**P75**
Panel A: Polymarket terminal ($N = 19{,}649$)					
Prediction market probability	0.3132	0.1450	0.3499	0.0375	0.5600
Benchmark probability	0.3056	0.1404	0.3428	0.0466	0.5242
Signed mispricing	0.0076	0.0028	0.0398	-0.0090	0.0187
Absolute mispricing	0.0221	0.0132	0.0340	0.0052	0.0281
Maturity (days)	3.6941	3.6666	2.0502	1.7081	5.6665
DVOL (annualised %)	44.3301	44.1100	7.3247	38.0300	48.7800
Panel B: Kalshi 30-minute baseline ($N = 3{,}135$)					
Prediction market probability	0.1847	0.1500	0.1693	0.0500	0.2700
Benchmark probability	0.1216	0.1216	0.0783	0.0603	0.1892
Signed mispricing	0.0631	0.0199	0.1477	-0.0134	0.0978
Absolute mispricing	0.0886	0.0457	0.1339	0.0170	0.1042
Maturity (days)	0.0363	0.0385	0.0052	0.0337	0.0402
DVOL (annualised %)	45.8272	44.8800	6.7572	39.7300	51.7600

Table 3: Observation-level descriptive statistics for the main empirical samples. Prediction market and benchmark probabilities are expressed on a zero-to-one scale. Signed mispricing equals the prediction market probability minus the DVOL-based benchmark probability; absolute mispricing is its absolute value.

In the Polymarket terminal sample, the mean and median of signed mispricing are close to zero, while absolute deviations remain relatively contained at the observation level. Kalshi has higher mean signed and absolute mispricing, together with substantially greater dispersion. The maturity statistics also illustrate the limited comparability of the two samples: the average Polymarket terminal horizon is approximately 3.7 days, compared with about 52 minutes for Kalshi. Average DVOL is broadly similar across the two samples.

Figure 1 compares the observation-level distributions of signed mispricing in the Polymarket terminal sample and the Kalshi 30-minute baseline. The horizontal axis is restricted to $[-0.30, 0.30]$ to keep the central part of each distribution visible.

Both samples have substantial probability mass near zero, but the Polymarket terminal distribution is more tightly concentrated. The Kalshi distribution is more dispersed and has a more pronounced positive tail. This visual comparison is descriptive and should

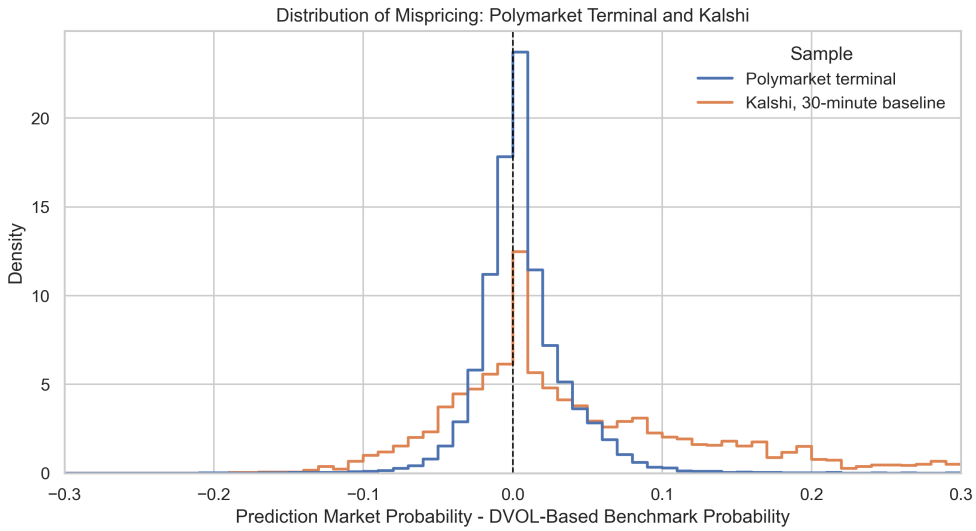


Figure 1: Distribution of signed mispricing in the Polymarket terminal sample and the Kalshi 30-minute baseline. The horizontal axis is restricted to $[-0.30, 0.30]$.

be interpreted in light of the differences in maturity, contract design, and institutional setting documented above.

Figure 2 turns to payoff structure within Polymarket by comparing terminal and path-dependent contracts. The same horizontal restriction is applied.

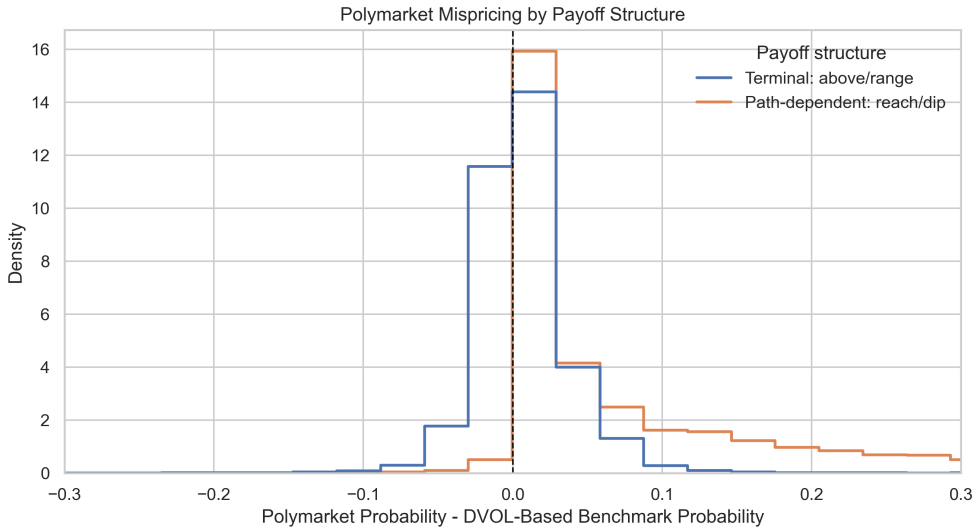


Figure 2: Distribution of Polymarket signed mispricing by payoff structure. Terminal contracts include *above* and *range* markets, while path-dependent contracts include *reach* and *dip* markets. The horizontal axis is restricted to $[-0.30, 0.30]$.

Terminal-contract mispricing is more tightly concentrated around zero. The path-dependent distribution has a heavier positive tail and contains more large positive observations. These differences must be interpreted cautiously because the terminal benchmark is not tailored to the payoff structure of path-dependent contracts.

6.3 Main Result: Polymarket Terminal Contracts

The main sample contains 19,649 observations from 2,847 Polymarket terminal markets. Mean mispricing at the observation level is 0.007551. Markets contribute different numbers of price observations, however, and an observation-weighted mean gives more influence to frequently observed markets. The preferred estimate therefore averages mispricing within each market before assigning equal weight to the resulting market-level means.

This procedure produces an average mispricing of 0.007496, or approximately 0.75 percentage points. Its 95% market-level bootstrap confidence interval, [0.006709, 0.008285], lies entirely above zero.

Table 4 reports the baseline estimate together with the seven-day maturity restriction and the filtered monthly result.

Sample	**Mean mispricing**	**95% confidence interval**
Polymarket terminal, market level	0.007496	[0.006709, 0.008285]
Polymarket terminal, $\tau \leq 7$ days	0.007483	[0.006704, 0.008271]
Polymarket terminal, filtered monthly	0.006445	[0.004252, 0.008637]

Table 4: Main Polymarket terminal estimate and robustness checks. The first two confidence intervals are market-level bootstrap intervals. The filtered monthly interval is a t-based interval calculated across monthly averages.

6.4 Supplementary Evidence from Kalshi

The Kalshi analysis uses timestamped transaction prices. The 30-minute baseline contains 3,135 contracts from 485 daily events and selects the first valid transaction recorded within 30 minutes of market opening.

Across individual contracts, mean mispricing is 0.063073. When mispricing is first averaged within each daily event and each event receives equal weight, the estimate rises to 0.070670, or approximately 7.1 percentage points. The corresponding 95% event-level bootstrap confidence interval is [0.064565, 0.077081].

The narrower trade-selection window gives a similar result. The 15-minute sample contains 2,554 contracts from 483 events and has an event-level mean mispricing of 0.076731. Its 95% confidence interval is [0.070557, 0.083126].

Table 5 presents both estimates.

Kalshi sample	Contracts	Events	Event-level mean and 95% CI
First trade within 30 minutes	3,135	485	0.070670 [0.064565, 0.077081]
First trade within 15 minutes	2,554	483	0.076731 [0.070557, 0.083126]

Table 5: Kalshi baseline result and timing robustness check.

Average deviations are larger in Kalshi than in the main Polymarket sample, but this difference should not be read as a direct ranking of market efficiency. Kalshi contracts have shorter horizons, use mutually exclusive price bins, and are observed through the first transaction after opening. The platform also operates under a different institutional and regulatory structure. The Kalshi estimates are consequently interpreted as supplementary evidence of cross-market differences, not as a one-to-one comparison with Polymarket terminal contracts.

6.5 Path-Dependent Polymarket Contracts

The Polymarket path-dependent sample contains 10,323 observations from 584 markets. Its observation-weighted mean mispricing is 0.084643. Averaging first within each market and assigning equal weight across markets produces a mean of 0.076566, with a 95% bootstrap confidence interval of [0.067337, 0.085983].

This estimate is considerably larger than the corresponding result for terminal contracts. The two estimates do not refer to theoretically equivalent claims, however. A *reach* or *dip* contract depends on whether Bitcoin touches a threshold before expiration, while the DVOL-based proxy calculates the probability of a terminal outcome.

The contrast illustrates the importance of payoff classification in the empirical design. Applying a terminal probability proxy to a path-dependent claim produces a comparison that must be interpreted in light of the mismatch between the payoff and the benchmark. For this reason, *reach* and *dip* contracts are excluded from the principal interpretation and retained as a separate comparison sample.

6.6 Regression Evidence

The regression analysis considers the conditional relationships between signed mispricing and benchmark probability, time to maturity, DVOL-based volatility, contract type, and market environment. The preferred specifications use clustered standard errors.

In the Polymarket terminal sample, the coefficients on benchmark probability, time to maturity, DVOL-based volatility, and the range-contract indicator are positive and statistically significant at the 1% level. The omitted contract category is *above*, and the

standard errors are clustered by Polymarket market.

The unified specification uses Kalshi as the reference market. Conditional on the reported controls, Polymarket terminal observations are associated with lower signed mispricing than Kalshi observations. The positive path-dependent coefficient represents an additional difference relative to Polymarket terminal contracts.

Benchmark probability also has a positive coefficient in the unified model, although its p-value of approximately 0.052 places it just outside the conventional 5% significance level. Time to maturity has a positive and statistically significant coefficient. By contrast, the DVOL coefficient is not statistically distinguishable from zero.

Because benchmark probability also enters the definition of signed mispricing, its coefficient should be interpreted with particular caution. Part of the estimated association is mechanical, and the coefficient is not treated as an independent structural determinant of mispricing.

Table 6 reports the selected coefficients and cluster-robust standard errors.

Variable	**Polymarket terminal**	**Unified sample**
DVOL-based benchmark probability	0.0204^{***}	0.0276^{*}
	(0.0014)	(0.0142)
Time to maturity, days	0.0019^{***}	0.0008^{***}
	(0.0002)	(0.0002)
DVOL-based volatility	0.0231^{***}	-0.0100
	(0.0055)	(0.0346)
Range contract indicator	0.0075^{***}	–
	(0.0009)	
Polymarket market indicator	–	-0.0638^{***}
		(0.0047)
Path-dependent indicator	–	0.0409^{***}
		(0.0085)
Observations	19,649	33,107
R^2	0.0320	0.3053
Clustering unit	Polymarket market	Polymarket market / Kalshi event

Table 6: Selected coefficients from the preferred clustered regressions. The dependent variable is signed mispricing, defined as the prediction market probability minus the DVOL-based benchmark probability. Clustered standard errors are reported in parentheses. Significance levels are denoted by $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

These regressions describe conditional relationships within the constructed datasets and are not given a causal interpretation. Differences in horizon, payoff structure, liquidity, and market design may remain relevant even after conditioning on the reported variables.

6.7 Robustness Checks

The robustness analysis first considers alternative sample definitions and units of inference. Market-level bootstrap inference for Polymarket accounts for repeated observations and prevents markets with more recorded prices from receiving greater inferential weight. Restricting the terminal sample to observations with no more than seven days remaining until expiration leaves the estimated mean almost unchanged.

The monthly exercise moves the analysis to a more aggregated level. Mispricing is averaged within each calendar month for the Polymarket terminal sample and the Kalshi 30-minute baseline. Sample-months containing fewer than 100 observations are omitted from the figure.

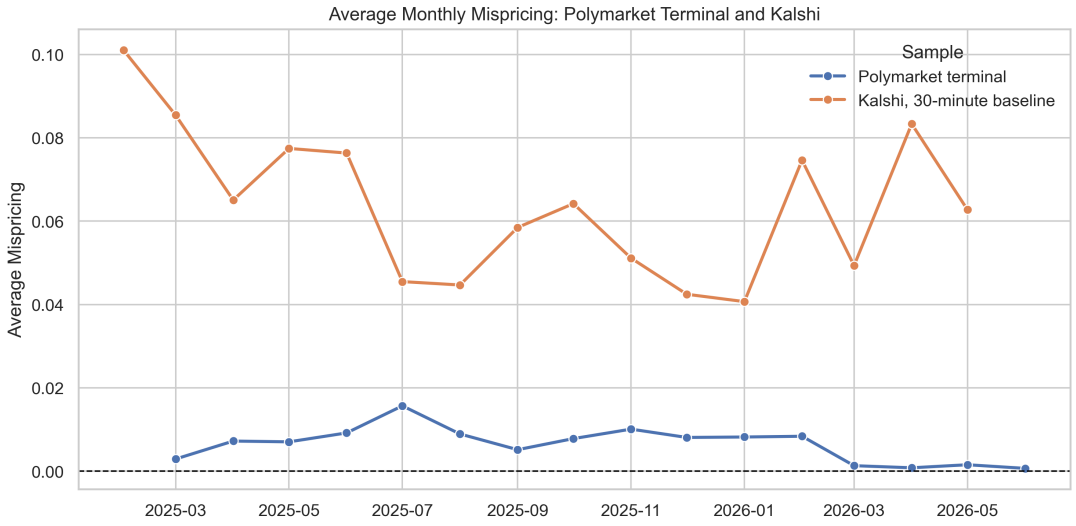


Figure 3: Average monthly mispricing for Polymarket terminal contracts and the Kalshi 30-minute baseline, excluding sample-months with fewer than 100 observations.

Figure 3 shows positive average mispricing in both samples, although its magnitude changes over time. Kalshi records larger monthly deviations, while the Polymarket terminal series remains closer to zero. The figure is descriptive; the formal monthly inference is conducted separately.

After applying the 100-observation threshold, 16 calendar months remain in the Polymarket terminal sample. Mean monthly mispricing is 0.006445. Treating the month as the unit of analysis, the two-sided 95% t-confidence interval calculated across the 16 monthly averages is $[0.004252,\ 0.008637]$.

The interval remains entirely above zero. The positive average difference is therefore also present when repeated observations are replaced by monthly averages and sparsely populated months are excluded.

The Kalshi timing rule is tested by replacing the 30-minute baseline with the first valid transaction recorded within 15 minutes of opening. As reported in Table 5, the narrower window produces a slightly larger estimate, while both confidence intervals remain entirely above zero.

Appendix A compares the preferred cluster-robust estimates with results based on HC3 heteroskedasticity-robust covariance estimators. The coefficient estimates are unchanged, but clustering increases several standard errors. In particular, benchmark probability is statistically significant in the unified HC3 specification but has a p-value of approximately 0.052 after clustering. The clustered specifications remain preferable because they account for repeated Polymarket observations and for dependence among Kalshi contracts belonging to the same event.

7 Discussion and Conclusion

This chapter interprets the empirical findings in light of the institutional differences across markets, the characteristics of the DVOL-based benchmark, and the related literature. It first discusses the main Polymarket result and the evidence from Kalshi and path-dependent contracts, before considering the contribution, limitations, and broader conclusions of the analysis.

7.1 Interpretation of the Main Findings

The main result is a positive average difference between Polymarket terminal probabilities and the DVOL-based benchmark. After each market is given equal weight, the estimated mean is 0.007496, or approximately 0.75 percentage points, and its market-level bootstrap confidence interval excludes zero.

The small signed average does not imply that every observation lies equally close to the benchmark. At the observation level, mean absolute mispricing is 0.0221, or about 2.21 percentage points, while median absolute mispricing is 0.0132. At the level of average signed differences, the result is consistent with relatively close, though incomplete, cross-market alignment.

The seven-day maturity restriction produces an almost identical estimate. Since it removes only 33 of the 19,649 terminal observations, this result mainly shows that the small longer-horizon tail of the sample does not drive the baseline estimate. The average difference also remains positive when the observations are aggregated by month.

A positive signed difference means that Polymarket terminal prices are, on average, above the probability proxy produced by the benchmark. It does not show that Polymarket probabilities are biased relative to the physical probability of the event. Prediction market prices may reflect risk preferences, funding constraints, liquidity, or investor demand, while the benchmark is based on a simplified model and an aggregate implied-volatility input. As discussed by Manski (2006) and Wolfers and Zitzewitz (2006), the mapping from a traded contract price to a real-world probability depends on the assumptions imposed on preferences and market structure.

The regressions provide a conditional description rather than a causal explanation. Within the Polymarket terminal sample, signed mispricing is positively associated with benchmark probability, maturity, DVOL, and the range-contract indicator. The benchmark-probability coefficient requires particular caution because the same probability also enters the definition of signed mispricing; part of the association is therefore mechanical.

In the unified specification, Polymarket terminal observations have lower signed mispricing than Kalshi observations after conditioning on the included variables. The positive

path-dependent coefficient captures an additional difference relative to Polymarket terminal contracts. These coefficients may absorb remaining differences in contract design, liquidity, and participant composition, so they should not be read as structural effects.

7.2 Institutional Setting and Payoff Comparability

Kalshi produces a much larger event-level estimate of approximately 7.1 percentage points. The 15-minute timing restriction gives a similar result, so the baseline estimate does not appear to be driven solely by transactions first recorded between 15 and 30 minutes after opening. Nevertheless, the comparison does not rank the two platforms by efficiency. Kalshi contracts have much shorter horizons, are organised as mutually exclusive price bins, and are represented by the first valid trade after opening. Their prices are also formed in a different regulatory and institutional environment.

The use of first trades introduces a further qualification. A timestamped transaction is preferable to a retrospectively updated quote because its information set can be identified, but the first trade may be noisy or occur in a thin market. Requiring a valid trade may also select contracts with greater trading activity. Event-level resampling accounts for the dependence among bins belonging to the same event, but it cannot remove these economic differences from the sample.

Payoff structure is equally important for the Polymarket comparison. The path-dependent sample has a market-level mean mispricing of 0.076566, far above the terminal estimate. A *reach* or *dip* contract, however, depends on the price path before expiration, whereas the benchmark evaluates a terminal event. The larger difference cannot be interpreted as evidence that path-dependent contracts are less efficient. It mainly shows that conclusions about cross-market pricing depend on whether the financial claims being compared have sufficiently similar payoffs.

7.3 Relation to the Literature and Contribution

The results are broadly consistent with the cross-market perspective of Fabi et al. (2026): prediction market and derivatives-based valuations can be closely related while systematic differences remain, and payoff structure affects how those differences should be read. A direct numerical comparison is not appropriate. Fabi et al. recover richer option-implied distributions and use valuation methods tailored to different payoff categories, whereas this thesis uses a parsimonious lognormal benchmark based on spot prices and DVOL.

The analysis also differs from Valery (2025). That study evaluates payoff relations and potential trading strategies across vanilla and binary options. The present thesis compares probability-like valuations and does not construct an executable portfolio. Consequently,

the reported mispricing measure should not be described as an arbitrage return, especially without bid–ask spreads, transaction costs, depth, and execution constraints.

The contribution is primarily empirical. The thesis constructs a transparent benchmark that respects information availability, separates terminal from path-dependent claims, and adjusts inference to the dependence structure of each dataset. It also adds timestamped Kalshi transactions as a supplementary regulated-market comparison. These choices make the scope of the evidence explicit and keep the central Polymarket result separate from comparisons involving less closely matched claims.

7.4 Limitations and Further Research

The main limitation concerns the benchmark. DVOL is a rolling 30-day annualised implied-volatility index, not a complete volatility surface. Using it in a lognormal terminal-price model imposes one volatility input across strikes and maturities and does not reproduce volatility skew, term structure, or the full risk-neutral distribution. The assumption of a zero risk-free rate is reasonable for the principal Polymarket terminal and Kalshi horizons, but it remains an approximation. These simplifications may matter most for tail, range, and path-dependent contracts.

Temporal alignment is conservative but not perfectly contemporaneous. Polymarket uses the latest daily inputs available at each observation, while Kalshi combines one-minute spot prices with the latest available daily DVOL value. This rule prevents look-ahead bias, but the volatility input may be stale for intraday trades. The two platform samples also cover different horizons and trading periods, limiting direct cross-platform interpretation.

Observed prices provide only a partial description of market conditions. The analysis does not control directly for bid–ask spreads, order-book depth, transaction costs, or trader composition. Nor does it compare the market probabilities with realised binary outcomes. It therefore studies cross-market pricing consistency rather than calibration, forecasting accuracy, or executable arbitrage.

Further work could construct maturity- and strike-specific probabilities from the full option surface and apply barrier-style models to path-dependent claims. Intraday implied-volatility inputs would improve the Kalshi alignment. Contract-level liquidity measures and executable price quotes would also allow a sharper distinction between valuation differences and tradable opportunities. A separate extension could test calibration against realised outcomes.

7.5 Conclusion

This thesis asks whether Bitcoin prediction market prices are consistent with probability benchmarks derived from the derivatives market. For Polymarket terminal contracts, average signed differences indicate relatively close, though incomplete, alignment. The preferred market-level estimate places prediction market probabilities about 0.75 percentage points above the DVOL-based benchmark. The estimate remains positive under monthly aggregation and is almost unchanged under the seven-day maturity restriction.

The supplementary evidence also shows why the definition of the claim matters. Kalshi records larger differences, but its contracts and trading environment are not directly comparable with the main Polymarket sample. Path-dependent Polymarket contracts likewise display larger deviations, although the terminal benchmark is not tailored to their payoff. Across the analysis, a measured difference is therefore treated as a gap between two market-based valuations, not as automatic evidence of irrationality, forecast failure, or arbitrage.

The broader conclusion is restrained. Bitcoin prediction markets and the derivatives-based benchmark convey related information, but they do not produce identical valuations. Part of the remaining gap may reflect institutional frictions or market segmentation; part may come from the benchmark itself. Distinguishing between these explanations requires richer option data and more detailed measures of trading conditions, but the evidence presented here provides a transparent basis for that comparison.

A Supplementary Regression Results

Table 7 examines the sensitivity of regression inference to the choice of covariance estimator. The underlying OLS specifications are identical to those reported in Table 6, so the estimated coefficients are unchanged. The comparison therefore focuses on the standard errors and p-values obtained under HC3 heteroskedasticity-robust and cluster-robust inference.

Variable	**Coefficient**	**HC3 SE**	**Clustered SE**	**HC3** p	**Clustered** p
Panel A: Polymarket terminal ($N = 19{,}649$, $R^2 = 0.0320$)					
DVOL-based benchmark probability	0.020425	0.000935	0.001392	< 0.001	< 0.001
Time to maturity, days	0.001887	0.000161	0.000183	< 0.001	< 0.001
DVOL-based volatility	0.023118	0.003628	0.005490	< 0.001	< 0.001
Range contract indicator	0.007546	0.000680	0.000892	< 0.001	< 0.001
Panel B: Unified sample ($N = 33{,}107$, $R^2 = 0.3053$)					
DVOL-based benchmark probability	0.027554	0.002948	0.014167	< 0.001	0.052
Time to maturity, days	0.000827	0.000016	0.000162	< 0.001	< 0.001
DVOL-based volatility	-0.010013	0.007975	0.034643	0.209	0.773
Polymarket market indicator	-0.063767	0.002654	0.004726	< 0.001	< 0.001
Path-dependent indicator	0.040947	0.001913	0.008520	< 0.001	< 0.001

Table 7: Sensitivity of regression inference to the covariance estimator. The dependent variable is signed mispricing. Panel A uses Polymarket market identifiers for clustered inference. Panel B uses hybrid clustering, with Polymarket market identifiers for Polymarket observations and Kalshi event identifiers for Kalshi observations. Reported p-values are two-sided.

The differences between the two covariance estimators are limited in the Polymarket terminal specification but are more pronounced in the unified sample. In particular, the standard error of the DVOL-based benchmark probability increases substantially under clustering, and its p-value rises from below 0.001 under HC3 to approximately 0.052. The Polymarket market indicator, the path-dependent indicator, and time to maturity remain statistically significant under both estimators, while DVOL remains statistically insignificant in the unified specification. Cluster-robust inference remains preferred because it accounts for dependence among observations from the same Polymarket market or Kalshi event.

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